ECG EASY JI

VIJAY BALAJI.K



"No doubt knowledge is valuable, but above it is Character"

Knowledge without character is dangerous!

OBJECTIVES

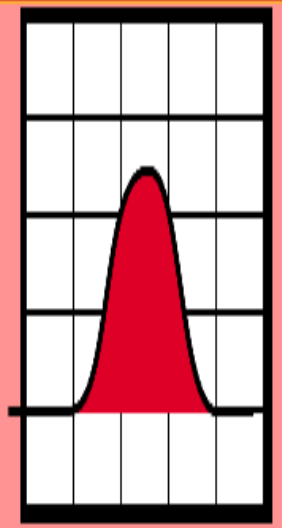
- NORMAL CONDUCTION SYSTEM
- R WAVE PROGRESSION
- ATRIAL ENLARGEMENT
- VENTRICULAR ENLARGEMENT
- SELF ASSESSMENT

CONDUCTION SYSTEM OF THE HEART

R WAVE PROGRESSION

NORMAL ATRIAL ACTIVITY

A normal P wave:

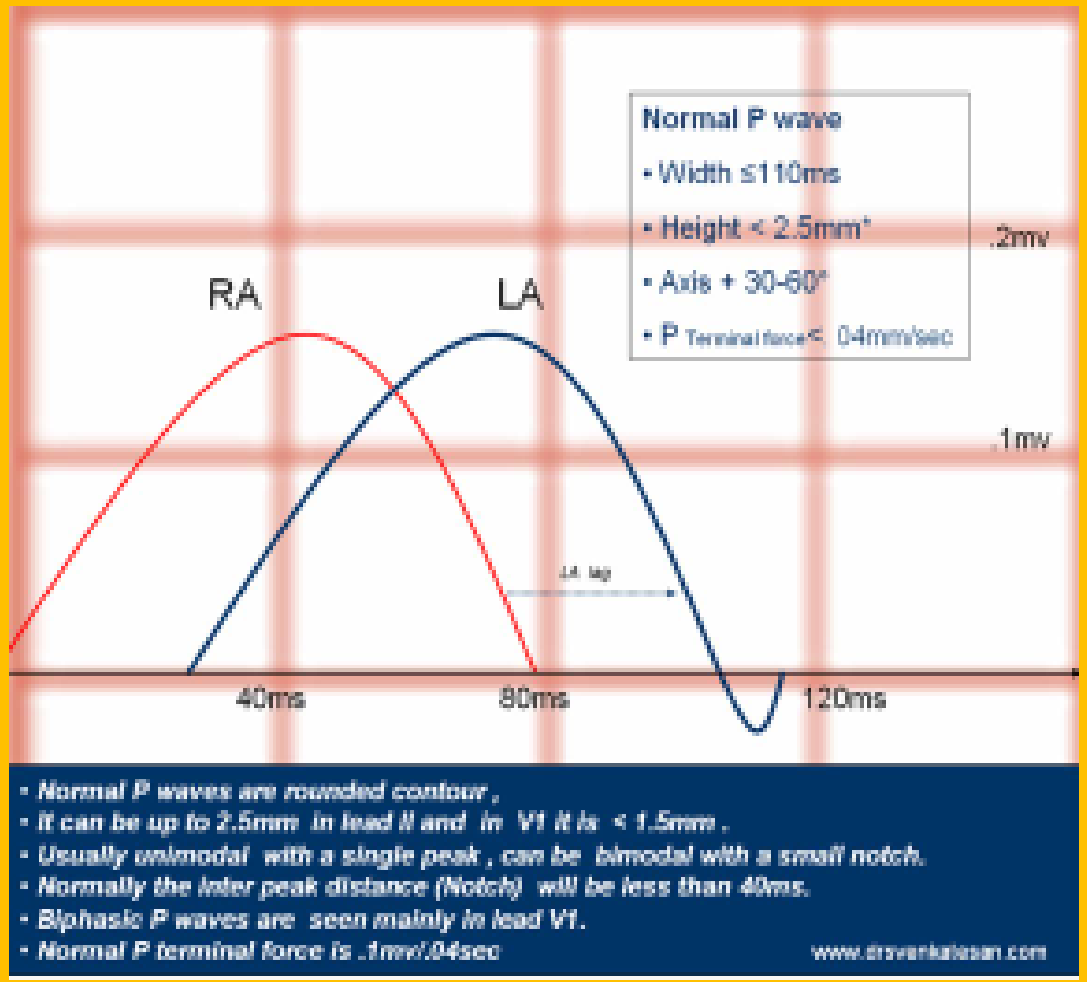


< 1.5 - 2.5

(chest leads) (limb leads

< 3

*small squares
at standard settings

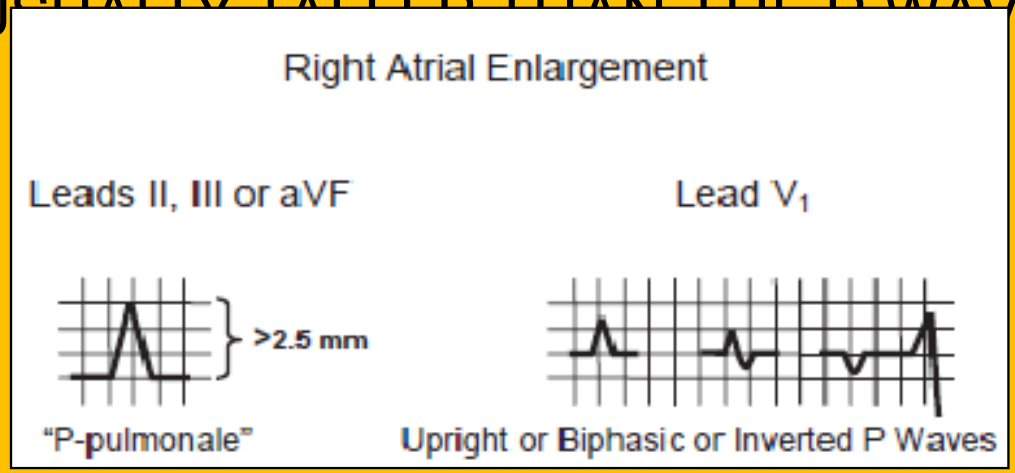


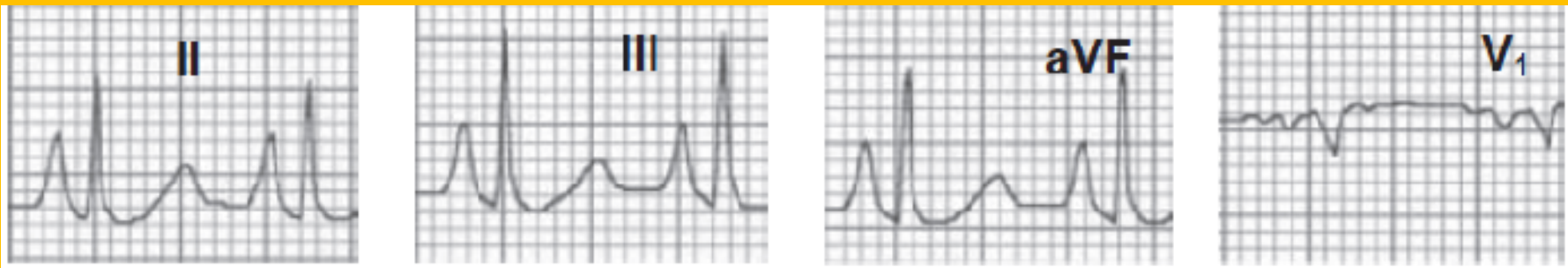
ATRIAL ENLARGEMENT

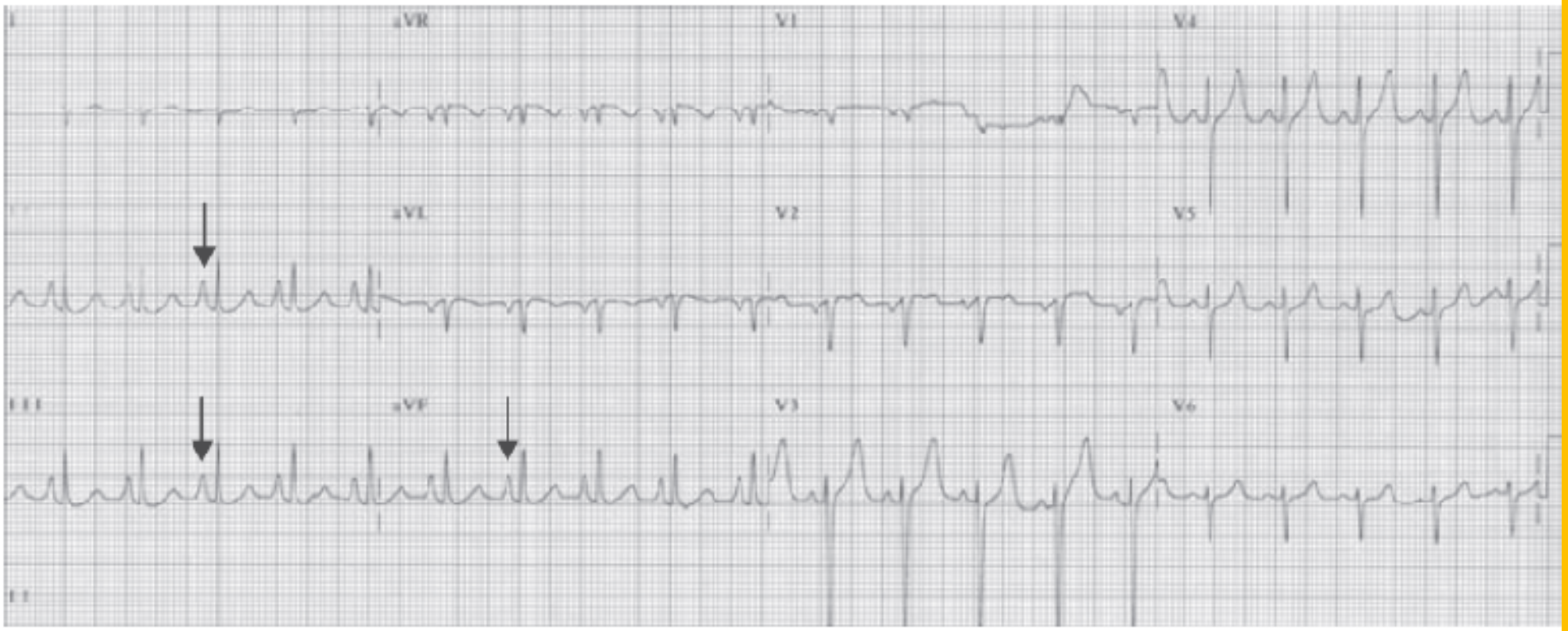
- NORMAL P WAVE IN LEAD 2 IS <2.5MM IN AMPLITUDE AND DURATION
- FIRST HALF OF THE P WAVE REPRESENT RIGHT ATRIAL ACTIVITY, SECOND HALF THE LEFT
- INCREASE IN AMPLITUDE >2.5MM = RIGHT ATRIAL ENLARGEMENT
- INCREASE IN DURATION >2.5MM = LEFT ATRIAL ENLARGEMENT (OFTEN SEEN AS A BIFID P WAVE IN LEAD 2)

RIGHT ATRIAL ENLARGEMENT

- THE RIGHT ATRIUM ENLARGES DOWNWARD AND TO THE RIGHT CAUSING THE DIRECTION OF THE SINUS IMPULSE TO SLIGHTLY SHIFT TO THE RIGHT OF 60.
- P WAVES ARE TALLER AND MORE PEAKED IN LEADS II, III, AND AVF.
- P WAVE IN LEAD III IS U[illegible]E IN LEAD I (P3>P1)

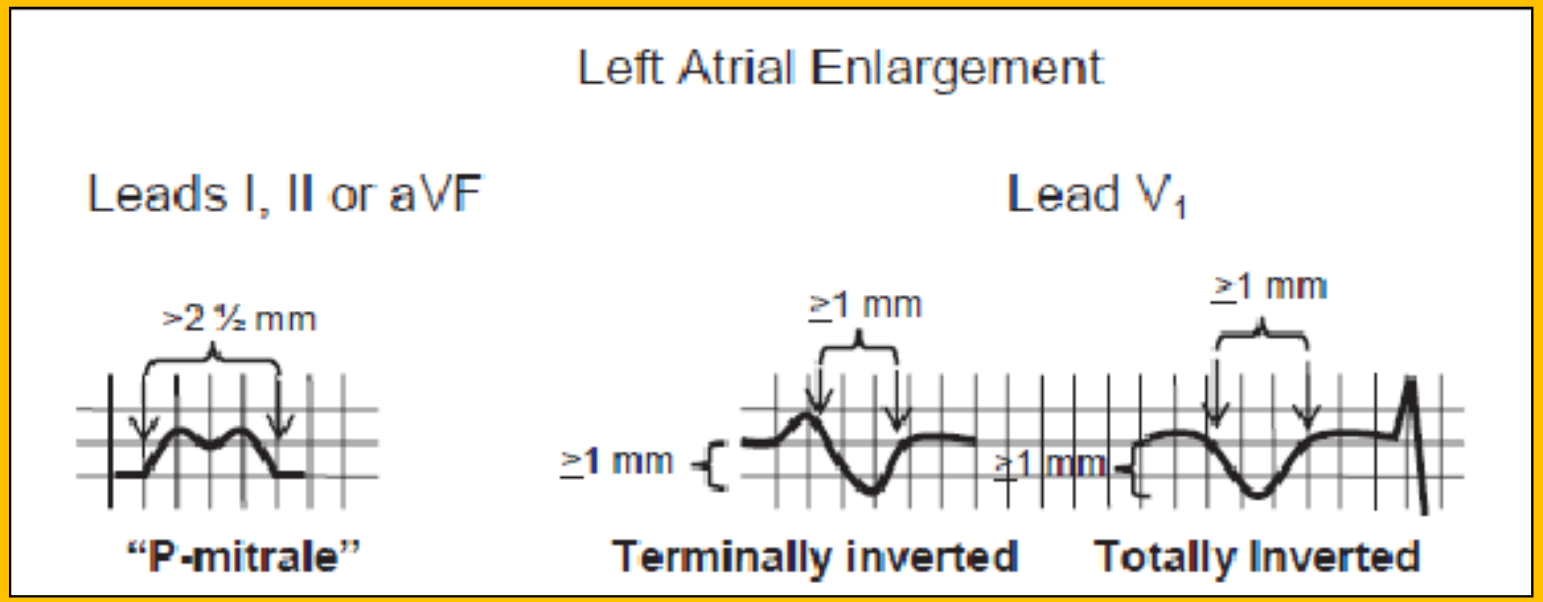



II
III
aVF
V1

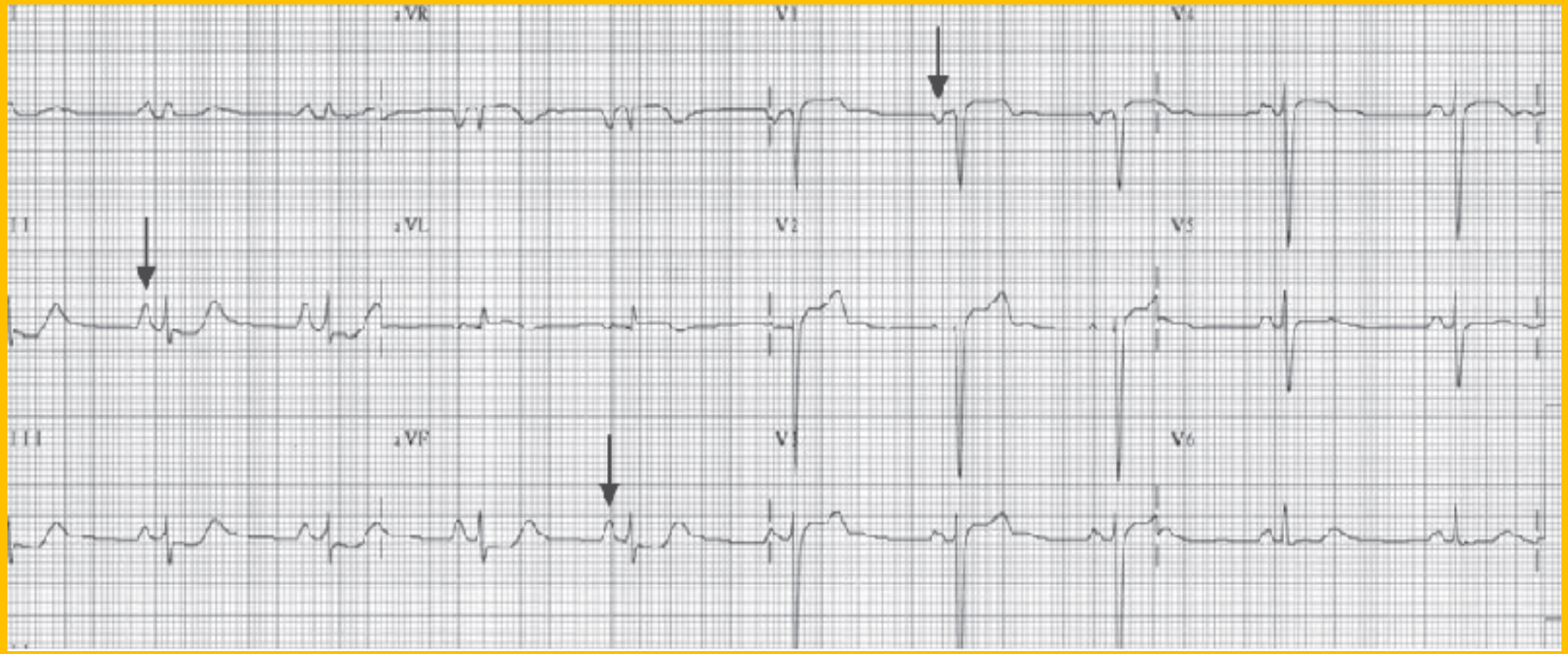


LEFT ATRIAL ENLARGEMENT

- THE LEFT ATRIUM ENLARGES TO THE LEFT AND POSTERIORLY SHIFTING THE P WAVE AXIS TO THE LEFT OF 45
- P WAVES ARE TALLER IN LEAD I THAN IN LEAD III (P1>P3).

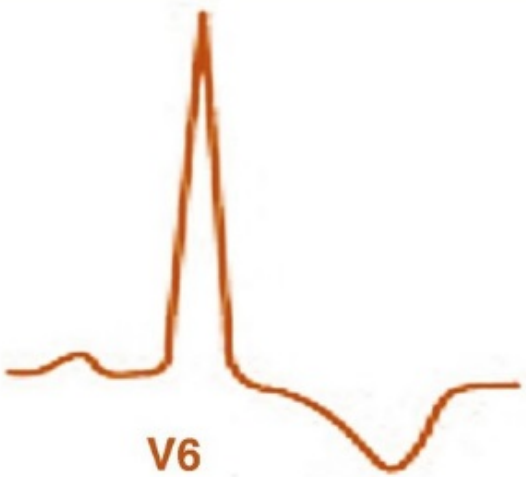


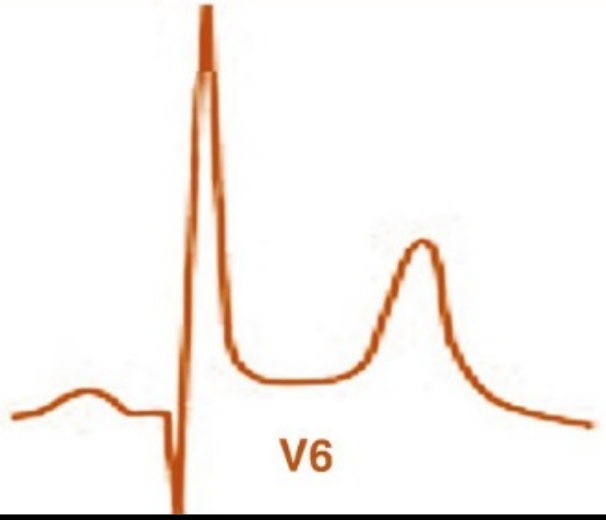
LAE Criteria	Sensitivity	specificity
Prolonged P wave duration > 120 msec in lead II	33	88
Prominent notching of P wave, usually most obvious in lead II, with interval between notches of 0.40 msec (P mitrale)	15	**100**
Ratio between the duration of the P wave in lead II and duration of the PR segment > 1.6 **Macruz index**	31	64
Increased duration and depth of terminal- negative portion of P wave in lead V_1 (P terminal force) so that area subtended by it > 0.04 mm-sec **Morris index**	69	93
Terminal negative deflection in V1 > 0.1 mV	60	93
duration > 0.04 s	**83**	80
Leftward shift of mean P wave axis to between −30 and −45 degrees		


I
aVR
V1
V4
II
aVL
V2
V5
III
aVF
V3
V6

VENTRICULAR ENLARGEMENT

Patterns of left ventricular Hypertrophy	
Pressure overload	**Volume overload**
Tall R wave with slurred ST segment and inverted T wave (strain pattern)	Tall R wave with upright T wave
	Deep Q wave
	Inverted U wave





CONCENTRIC LVH

↑ Global
LV Wall Thickness

Normal
LV Cavity Size

↑ LVM
↑ RWT

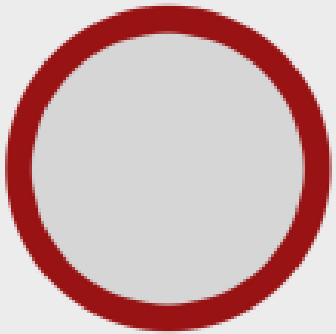


ECCENTRIC LVH

Normal Global
LV Wall Thickness

↑ LV
Cavity Size

↑ LVM
↓ RWT



CONCENTRIC REMODELING

↑ LV Wall
Thickness

Normal or Small
LV Cavity Size

Normal LVM
↑ RWT



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Mechanism of Q wave in volume over-load of LV

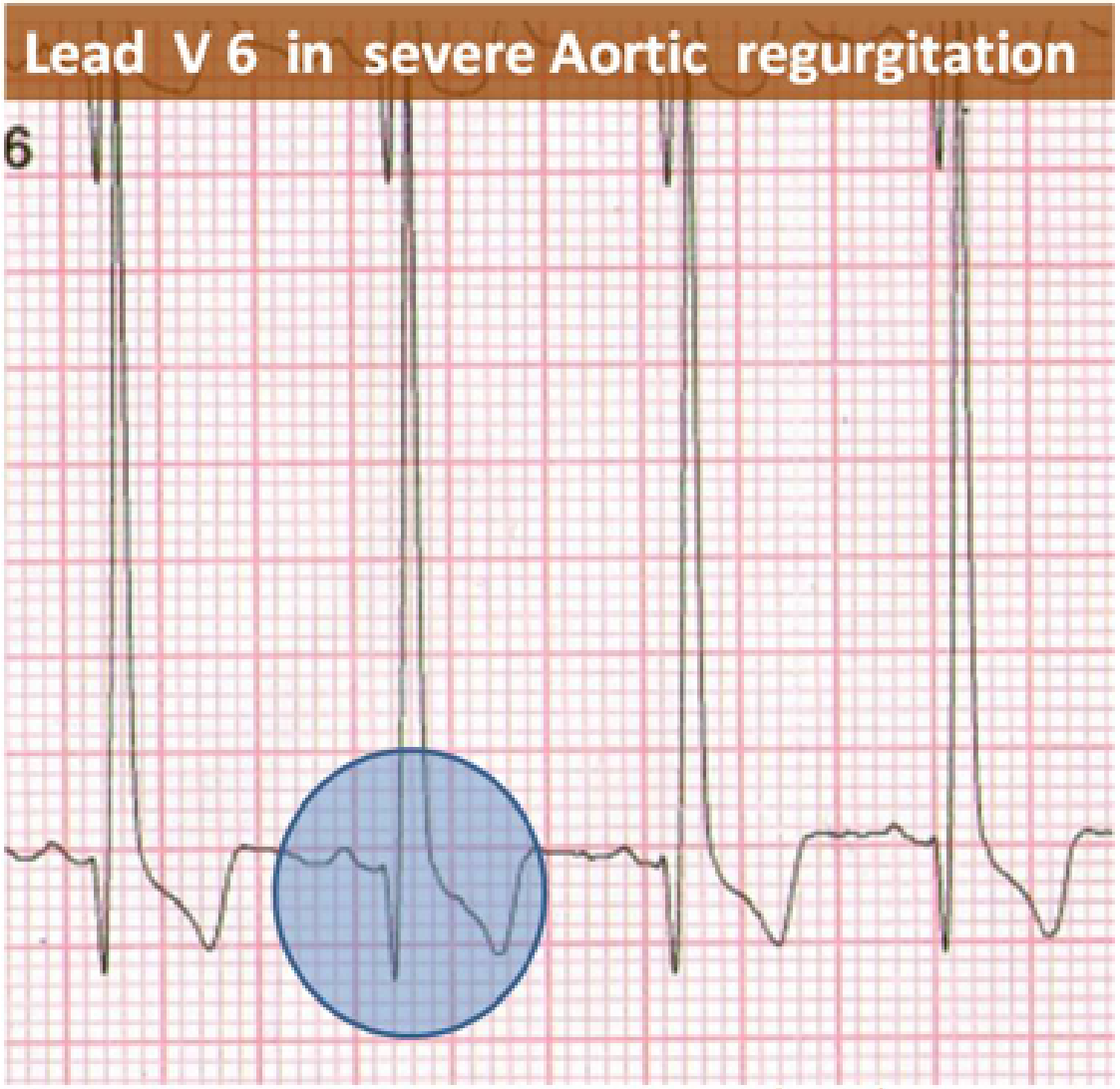
q represents LV end diastolic point .

At this point LV is maximally stretched , filled and stressed *

Maximum LVID corresponds to **q** wave

LV cavity along with mass of blood amplifies the **q** wave along with R wave (Brody effect).

* Please note : In LV systolic overload peak stress point is mid systole ie ST segment . So no Q waves .



RIGHT VENTRICULAR HYPERTROPHY

TABLE 12–1. Electrocardiographic Criteria of Right Ventricular Enlargement

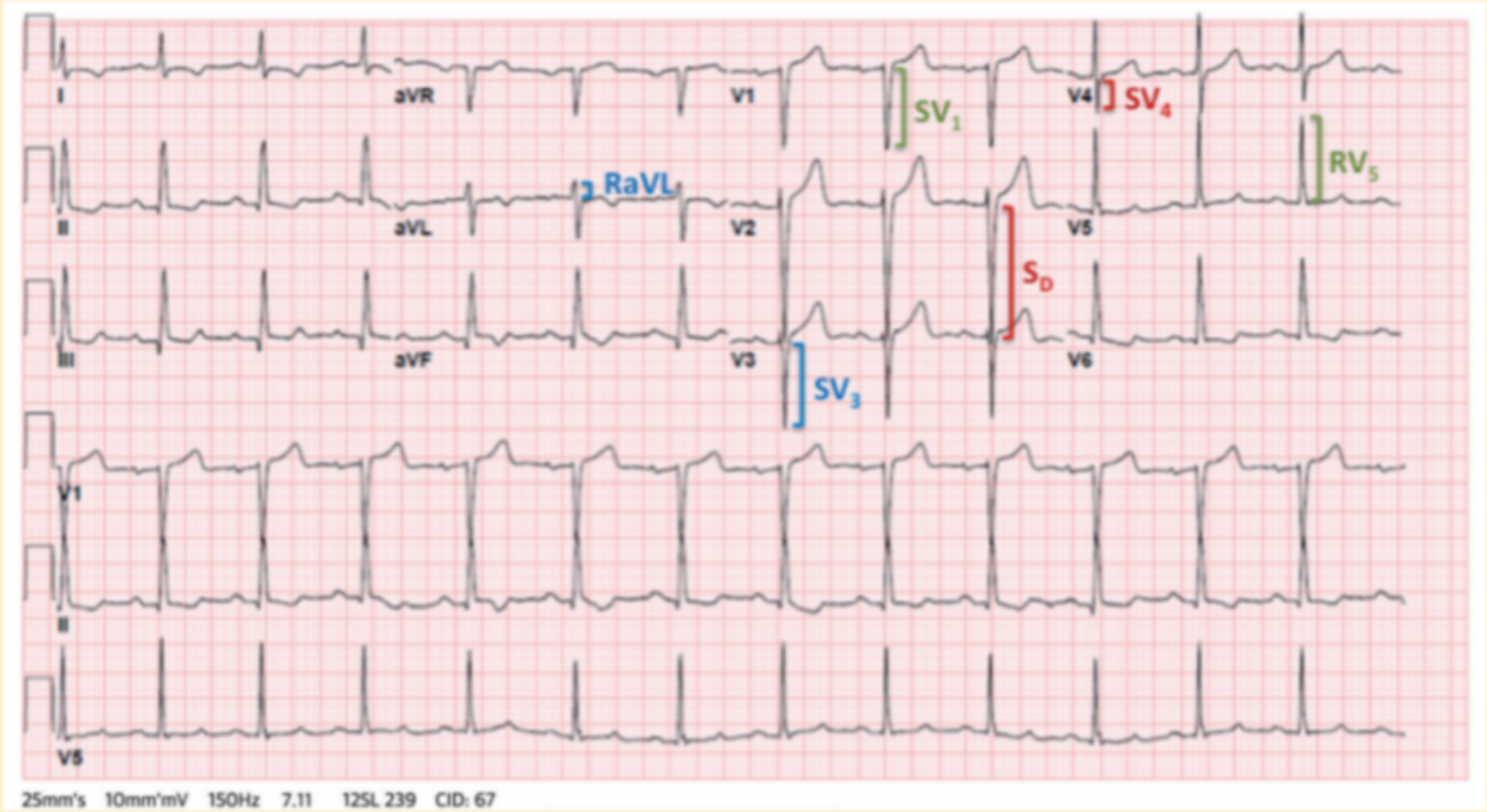
	Criterion	Sensitivity (%)	Specificity (%)
V_1	$R/S\ V_1 > 1$	6	98
	$R\ V_1 \geq 7$ mm	2	99
	qR in V_1	5	99
	S in $V_1 < 2$ mm	6	98
	IDT in $V_1 > 0.35$ s	8	98
V_5–V_6	$R/S\ V_5$–$V_6 \leq 1$	16	93
	$R\ V_5\ V_6 < 5$ mm	13	87
	$S\ V_5$-$V_6 \geq 7$ mm	26	90
$V_1 + V_6$	$HV_1 + SV_5$–$V_6 > 10.5$ mm	18	94
	ÂQRS ≥ 110°		
ÂQRS	S_I, S_{II}, S_{II}	15	96
		24	87

LEFT VENTRICULAR ENLARGEMENT

Voltage Criteria	Sensitivity (%)	Specificity (%)
1. $R_I + S_{III} > 25$ mm	10.6	100
2. RVI > 11 mm	11	100
3. RVL > 7.5 mm	22	95
4. $SV_1 + RV_5$-$V_6 \geq 35$ mm (Sokolow-Lyon)	22	100
5. RV_6 $V_6 > 26$ mm	25	98
6. RVI + $SV_3 > 28$ mm (men) or > 20 mm (women) (Cornell voltage criterion)	42	95
7. Cornell voltage duration measurement QRS duration × Cornell voltage > 2436 mm/seg	57	95
8. In V_1-V_6, the deepest S + the tallest R > 45 mm	45	93
9. Romhilt-Estes score > 4 points	55	85
10. Romhilt-Estes score > 5 points	35	95

In HTN : Sum of QRS voltage of 12 leads > mm.

Peguero–Lo Presti criteria for Diagnosis of Left Ventricular Hypertrophy



Deepest S wave in any lead and S wave in V4 [S_D + SV_4]

S_D + SV_4 ⩾2.3 mV for female subjects and ⩾2.8 mV for male = LVH

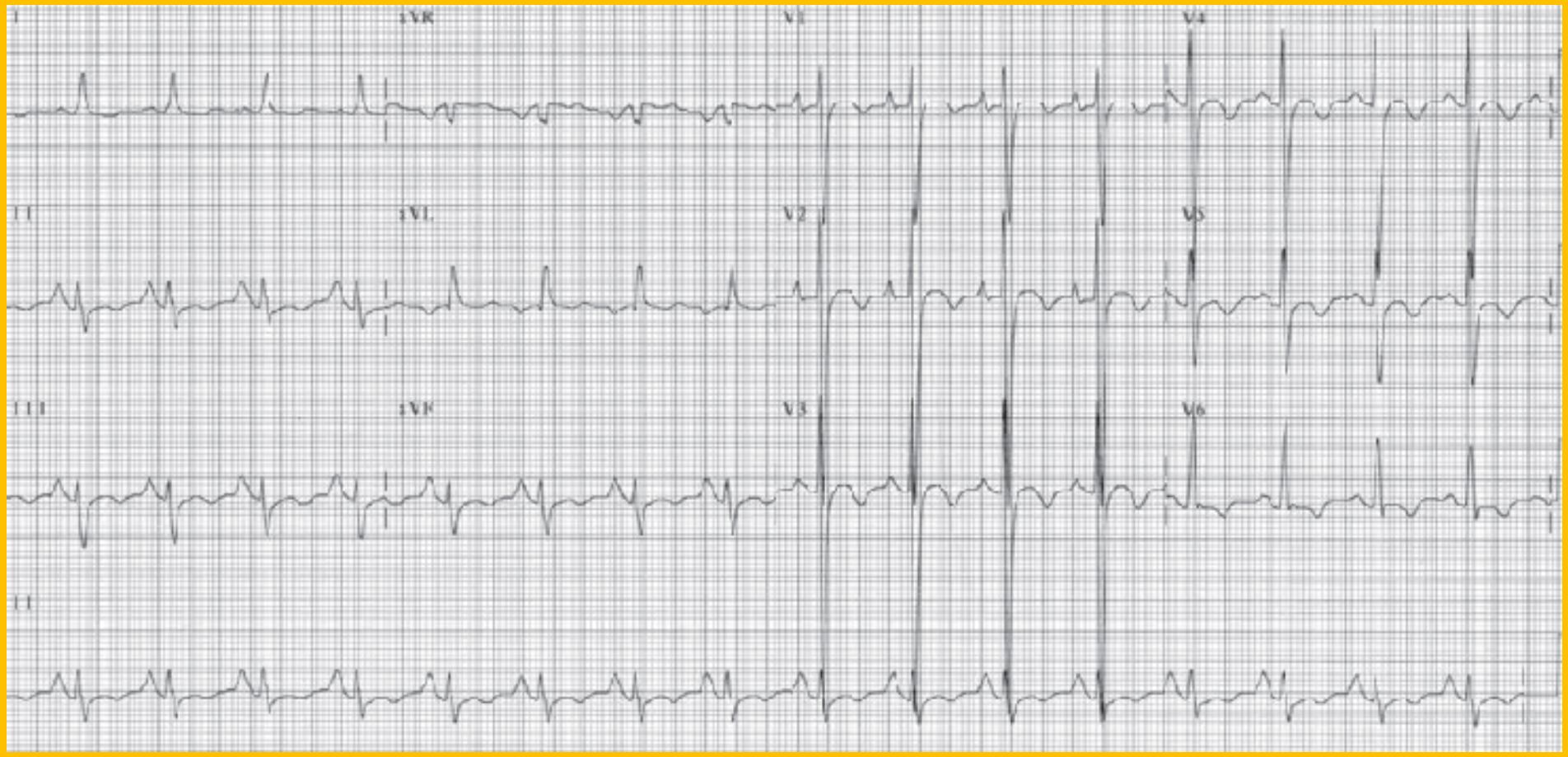
Table 13–4 Criteria for Diagnosis of Left Ventricular Hypertrophy in Presence of Complete Left Bundle-Brance Block		
	Sensitivity %	**Specificity %**
1. R in aVL ≥11 mm	24	100
2. Electrical axis ≥40 (or $S_2 \geq R_1$)	39	100
3. $SV_1 + RV_5$ or $RV_6 \geq 40$ mm	58	97
4. $SV_2 \geq 30$ mm and $SV_3 \geq 25$ mm	75	90

- *Hurst 12/e*

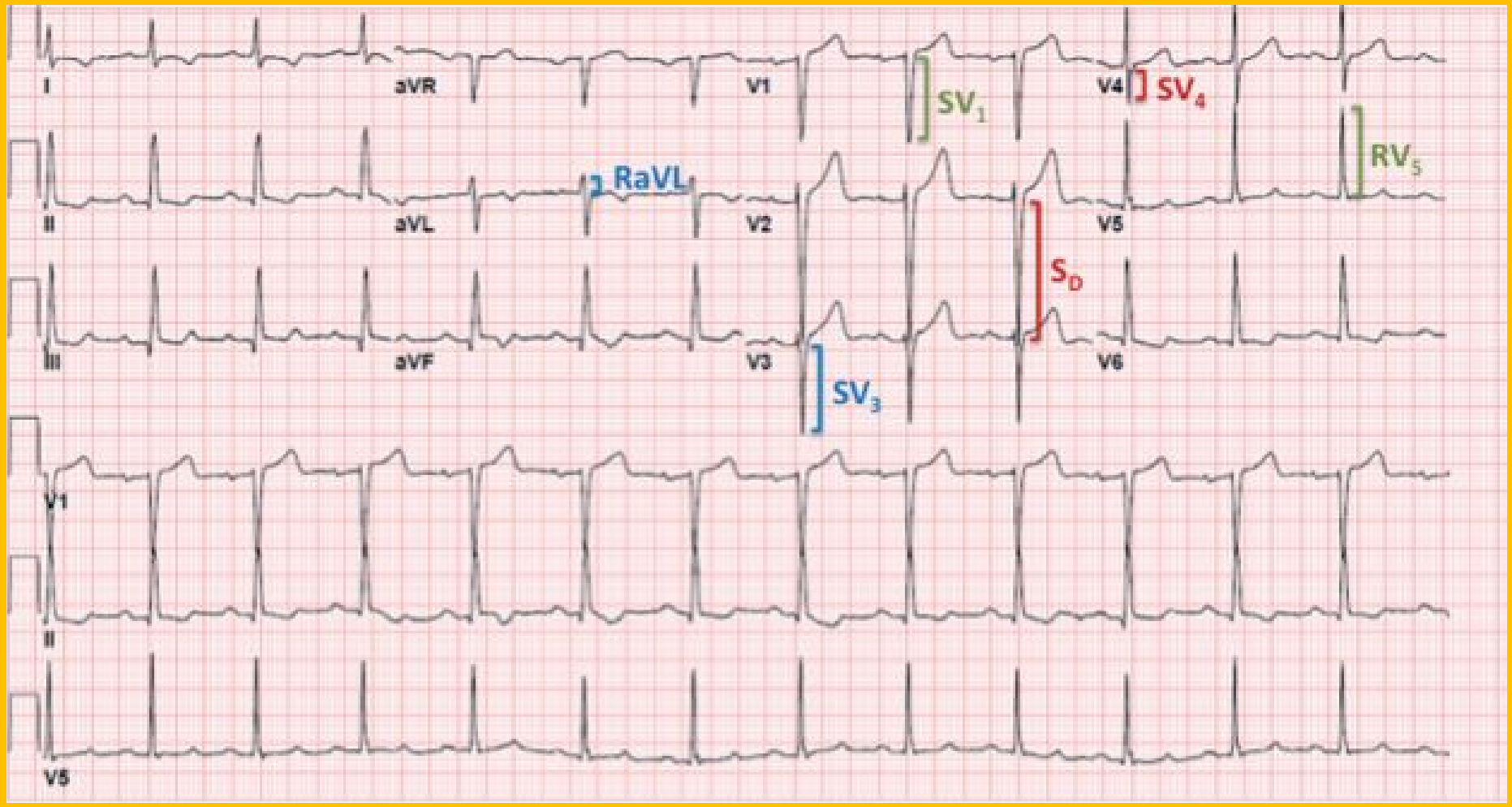
Table 13–5 Criterica for Diagnosis of Left Ventricular Hypertrophy in Presence of Left Anterior Fascicular Block[1]

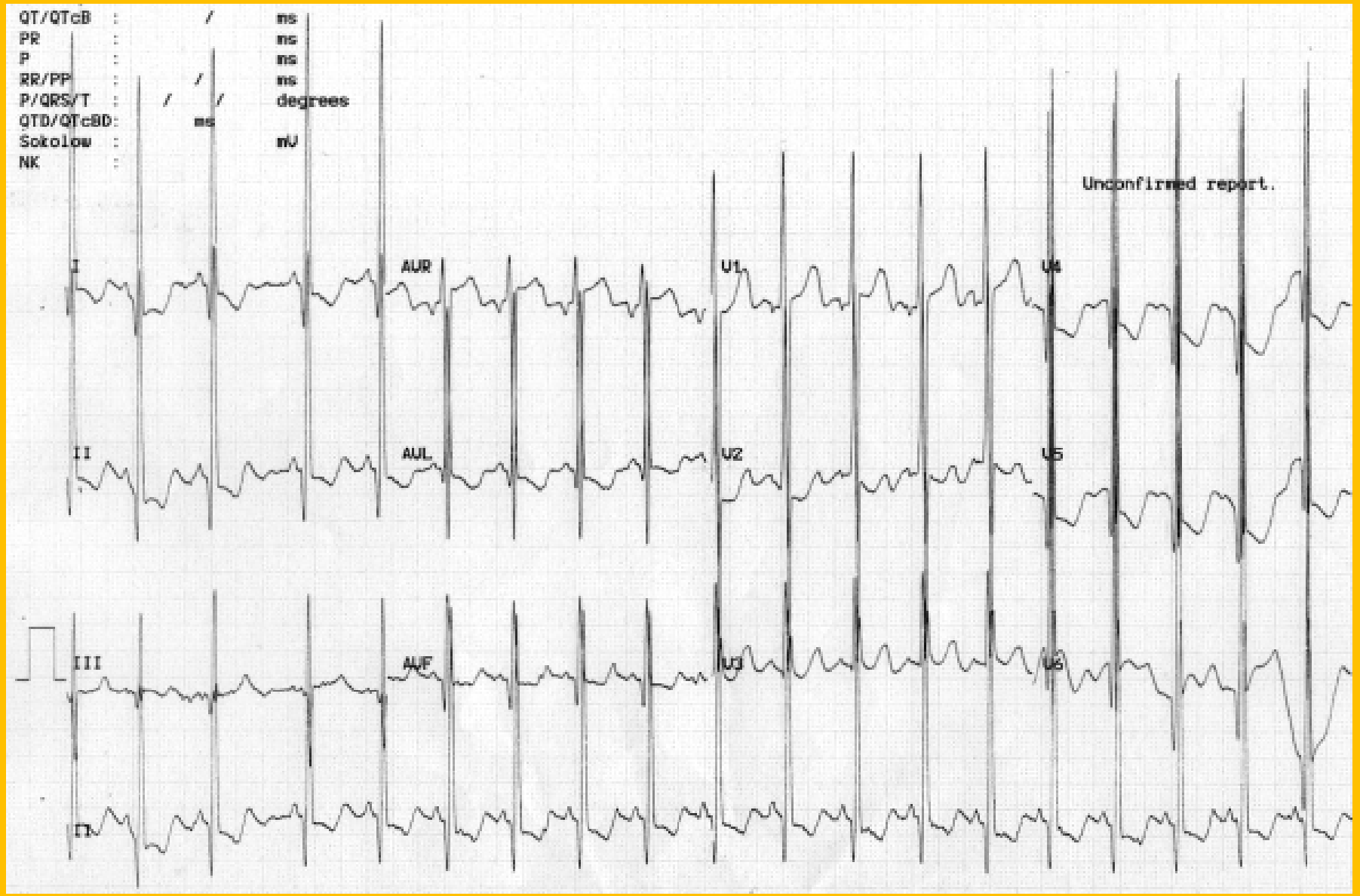
Study	ECG Criteria	Sensitivity (%)	Specificity (%)	Positive Predictive Value (%)	Negative Predictive Value (%)
Bozzi and Figini	$SV_1 + (RV_5 + SV_5) \geq 25$ mm	69	92	80	73
Milliken	RaVL ≥13 mm	35	92	82	56
Milliken	SIII ≥15 mm	38	87	77	57
Gerstch et al.[120]	SIII + maximal sum of R + S in any single precordial lead	96	87	89	95
Reevaluated Gerstch criteria		80	55	78	58

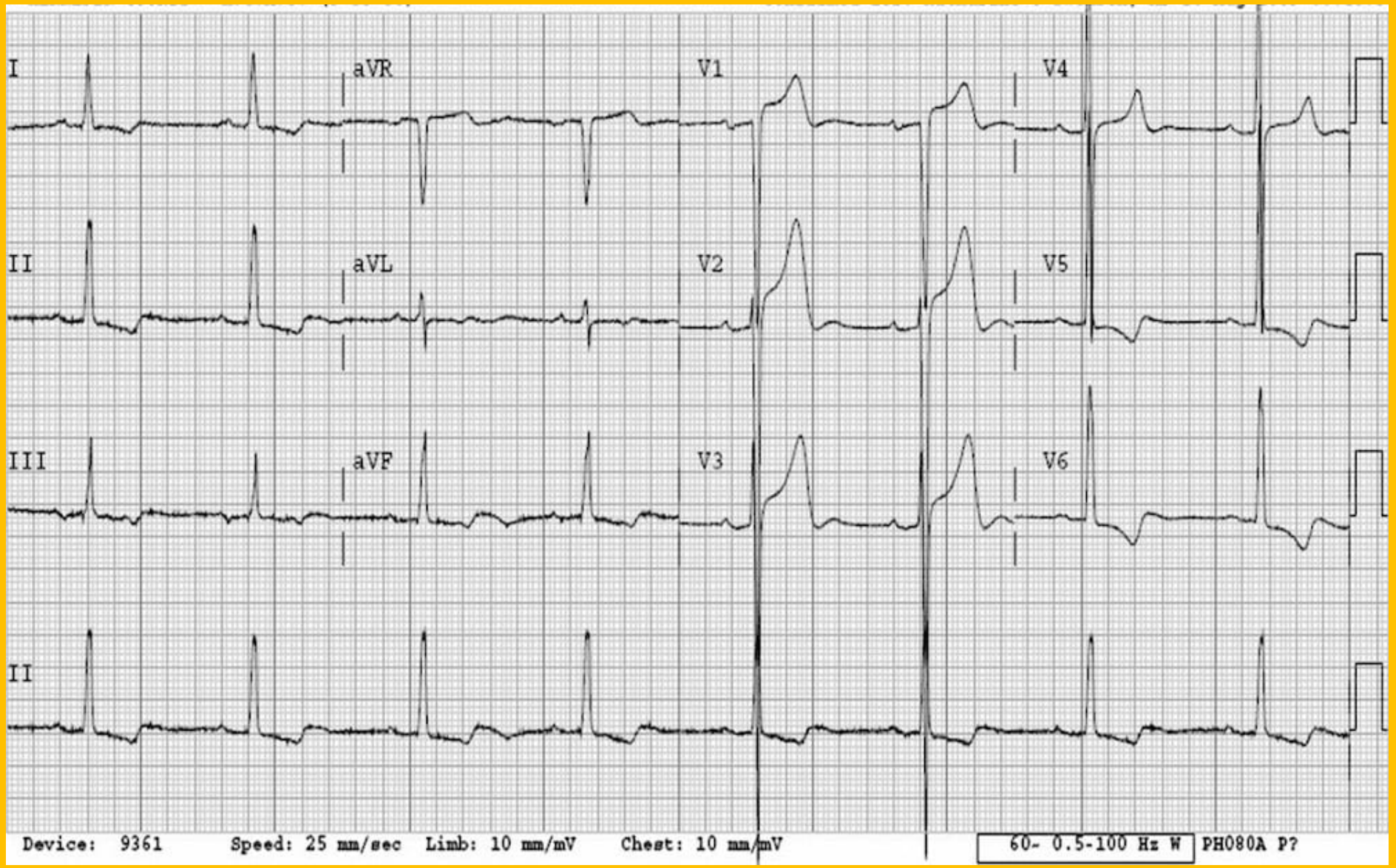
BIVENTRICULAR ENLARGEMENT – KATZ WACHTEL PHENOMENON

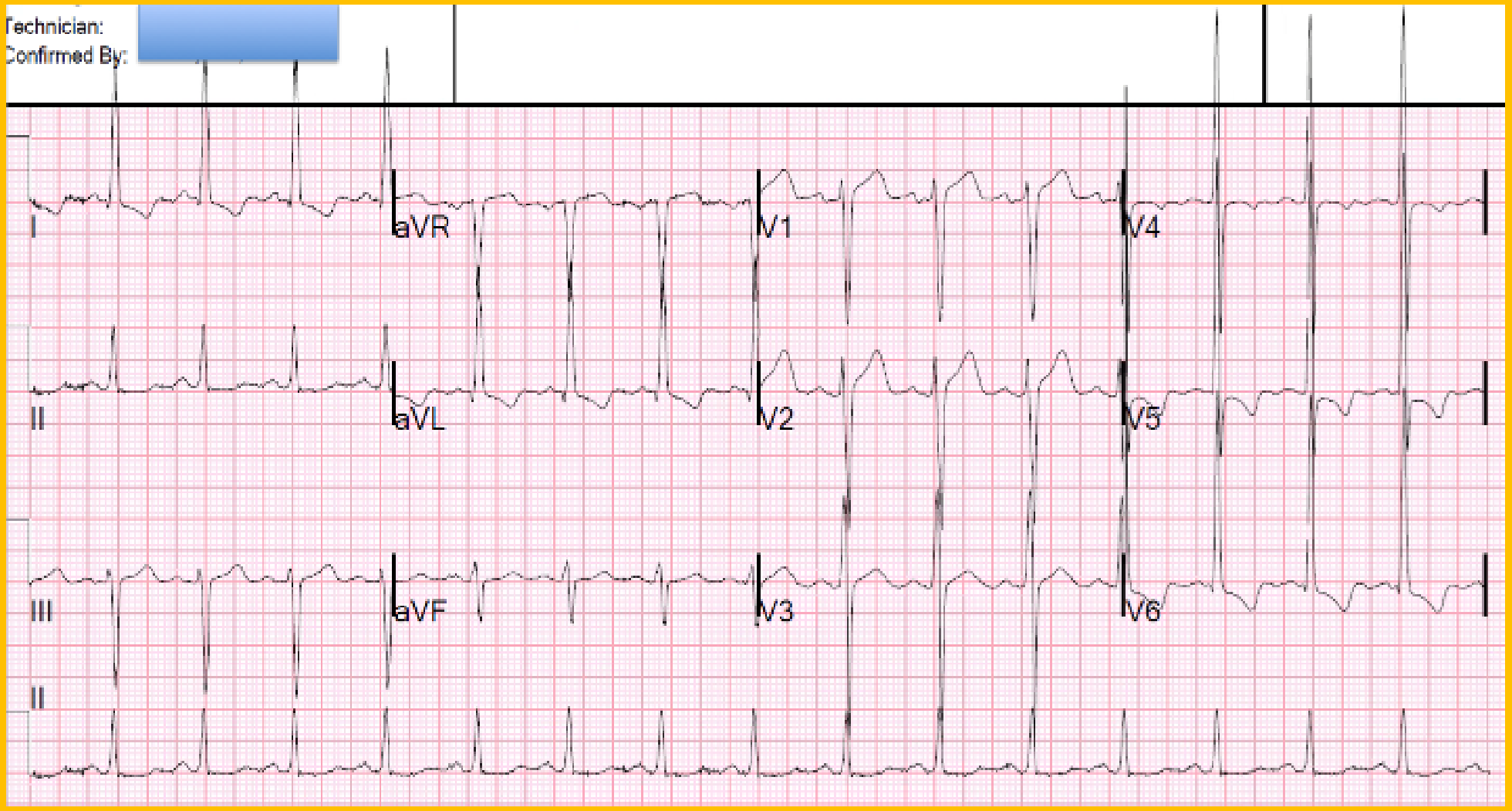


SELF ASSESSMENT




QT/QTcB : / ms
PR : ms
P : ms
RR/PP : / ms
P/QRS/T : / / degrees
QTD/QTcBD: ms
Sokolow : mV
NK :
Unconfirmed report.
I
AUR
V1
V4
II
AUL
V2
V5
III
AVF
V3
V6


I
aVR
V1
V4
II
aVL
V2
V5
III
aVF
V3
V6
II
Device: 9361
Speed: 25 mm/sec
Limb: 10 mm/mV
Chest: 10 mm/mV
60- 0.5-100 Hz W
PH080A P?


Technician:
Confirmed By:
I
aVR
V1
V4
II
aVL
V2
V5
III
aVF
V3
V6
II

QUERIES

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  - WHITE ARMY WHATSAPP GROUPS

PHONE NUMBER : ANY CORE MEMBERS
9611584473(PERSONAL)

IF YOU FIND ANY INTERESTING ECG FEEL FREE TO SEND IT TO THE ABOVE NUMBER



"Our character is but the stamp on our souls of the free choices of good and evil we have made through life.

-- John C. Geikie